

Sawera
22024

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Arithmetic Mean:

Def: The arithmetic mean is the sum of all values of data divided by the total number of values.

Formula:

$$\text{Mean} = \frac{\sum X}{N}$$

X = individual observation.

N = Total number of observation.

Characteristics:

Mathematically defined term, it's clearly defined by a formula.

Based on all the observation - every value in the dataset contribute to the mean.

Simple to understand easy to calculate.

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Algebraic property: The sum of deviation of all the observation from the mean is always Zero.

$$\Sigma(\bar{X} - x) = 0$$

Least Variation: The sum of squared deviations from the mean is minimum compared to the other value.

Symbols:

$$\text{Population mean} = \mu = \frac{\Sigma X}{N}$$

$$\text{Sample mean} = \bar{X} = \frac{\Sigma X}{n}$$

Example: Calculate the population mean:

Data: 2, 8, 10, 6, 12, 80

$$\mu = \frac{2+8+10+6+12+10}{6} = \frac{48}{6} = 8$$

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X	$X - \mu$
2	-6
8	0
10	2
6	-2
12	4
10	+2

$$\sum_i (X - \mu) = 0$$

Population Variance in Standard deviation:

Variance measure the average of the squared deviation from the mean. It represents how spread out the data values are:

$$\sigma^2 = \frac{\sum (X - \mu)^2}{N}$$

Standard deviation:

$$\sigma = \sqrt{\frac{\sum (X - \mu)^2}{N}}$$

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Example:

X	$X - \mu$	$(X - \mu)^2$
2	-6	36
8	0	0
10	2	4
6	-2	4
12	4	16
10	2	4

$$\sigma^2 = \frac{64}{6} = 10.66$$

$$\sigma = \sqrt{10.66} = 3.26..$$

Advantages:

- Uses all observations.
- Simple and widely used.
- Has defined algebraic properties
- Useful in advanced statistical analysis

Disadvantages:

Affected by extreme value.
Not suitable for qualitative data.
Cannot be determined for open-ended frequency distribution.

Median: Median is

defined as:

The middle value that divides the dataset into two equal halves when arranged in ascending order.

$$\text{Median} = \frac{\text{Middle Two values}}{2}$$

Example:

$$= 2, 6, 8, 12, 10, 10$$

$$= 2, 6, 8, 10, 10, 12$$

$$\text{Median} = \frac{8+10}{2} = 9$$

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$(X - \mu)$	(X)	$(X - \mu)$	$(X - \mu)^2$
$2 - 7 = -5$	2	$2 - 8 = -6$	36
$10 - 7 = 3$	10	$10 - 8 = 2$	4
$6 - 7 = -1$	6	$6 - 8 = -2$	4
$8 - 7 = 1$	8	$8 - 8 = 0$	0
$12 - 7 = 5$	12	$12 - 8 = 4$	16
$10 - 7 = 3$	10	$10 - 8 = 2$	4
$\sum X = 48$		$64 = \sum (X - \mu)^2$	
$N = 6$			

Population Variation = $\frac{\sum (X - \mu)^2}{N} = \frac{64}{6} = 10.667$

Standard deviation = $\sqrt{\frac{\sum (X - \mu)^2}{N}} = \sqrt{\frac{64}{6}}$

$\sqrt{10.66} = 5.33$